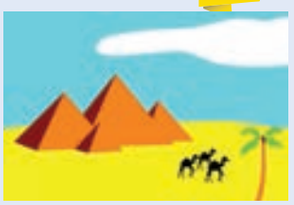
Obviously now, we see that

Circumference, $C = \pi d$ and $d = 2r$.

Thus for any circle with a given 'r' or 'd', we can find 'C' and vice-versa.

DO YOU KNOW?

1. Calculation of decimal digits of π is an interesting task among mathematicians.
2. The constant π was used in the construction of popular pyramids of Egypt.
3. Computers helped mathematicians to calculate the value of π with more than 12 trillion decimal digits.



Think

A circle has the shortest perimeter of all closed figures with the same area. Justify with an example.

Example 2.1 Calculate the circumference of the bangle shown in Fig. 2.5 (Take $\pi = 3.14$).

Solution

Given, $d = 6\ cm$, $d = 2r = 6\ cm$, $r = 3\ cm$

Circumference of a circle $= 2\pi r\ units$

$= 2\pi \times 3$

$= 18.84$

The circumference is 18.84 *cm*.

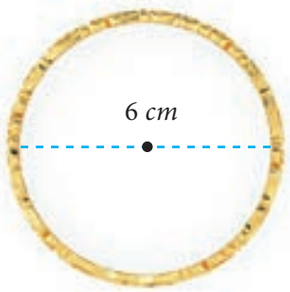


Fig. 2.5

Example 2.2 What is the circumference of the circular disc of radius 14 *cm*? (use $\pi = \frac{22}{7}$)

Solution

Radius of circular disc $(r) = 14\ cm$

Circumference of the disc $= 2\pi r\ units$

$= 2 \times \frac{22}{7} \times 14$

$= 88\ cm$

Example 2.3 If the circumference of the circle is 132 *m*. Then calculate the radius and diameter (Take $\pi = \frac{22}{7}$).

Solution

Circumference of the circle, $C = 2\pi r\ units$

The circumference of the given circle = 132 *m*

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